

Does the Assistant Check the Record? Knowledge-Base Poisoning and System-of-Record Validation in Utility Operations RAG

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Abstract: Retrieval-augmented generation (RAG) enables document-grounded technical lookup, but indexed repositories may accept low-trust contributions. We study targeted numeric answer-integrity corruption in utility-operations RAG with GridPoison-100: 35 source-grounded public items, 45 fictional asset-register items, and 20 legitimate-conflict items. The two pre-specified primary tests were null: fixed-context forced exposure produced 0% attacker-value adoption in both tiers, and a keyed system-of-record (SoR) check at 70% coverage did not significantly outperform a cautious prompt or isolate-then-aggregate voting (all Holm-adjusted $p = 1.0$). In post-hoc descriptive analysis under a base prompt that returns `VALUE: UNKNOWN` on unresolved conflict, one poison at $k = 5$ reached 70.0% of public-item queries; Tier 1 correct-response rate (CRR) fell from 75.7–78.6% to 8.6–12.9%, mostly through non-commitment, and conditional adoption rate (CAR) among exposed trials was 4.1%. In the pre-specified secondary $k = 2$ condition, CAR was 34.6% on the same 9 of 26 exposed trials for all three models. A locked commit-required control supported material prompt mediation for Mistral and partial mediation for Luna: CAR rose to 49.0% and 16.3%, respectively. In forced two-passage contexts, the poison’s clean twin suppressed adoption and a lone poison was always adopted, but a different clean support left 17 of 80 adoptions for Mistral, falsifying the pre-stated strong composition hypothesis. D3 is an upper-bound keyed record check: full coverage removed observed out-of-range commitments by construction but flagged 64.1–65.7% of otherwise-correct legitimate-conflict answers because 13 of 20 conflict records held generic profiles.

Index Terms: retrieval-augmented generation, knowledge-base poisoning, power systems, utility operations, large language models, data integrity

I. INTRODUCTION

Large language models are increasingly proposed as natural-language interfaces to technical documentation, including power-system applications [1], [2]. In power and utility settings, the most plausible near-term use is not autonomous control but constrained lookup: a field worker asks for an approach distance, a protection engineer checks an interconnection boundary, or a planner retrieves an approved asset limit. RAG makes such answers auditable by grounding them in external documents, yet it also creates a write-path dependency. If an indexed repository accepts contributions from contractors, shared drives, wikis, or externally supplied bulletins, a legitimate user can receive an incorrect value even when the user’s prompt is benign.

Knowledge-base poisoning of RAG is already established [3], [4]. Accordingly, our contribution is not another demonstration that injected text can influence an answer. Instead, we measure how a conflict-aware utility assistant separates silent corruption from non-commitment, and how a separate structured record trades filtering benefit against false flags on legitimate jurisdiction- and interconnection-specific answers.

We study three research questions. **RQ1** asks how targeted poisoning decomposes into retrieval exposure and conditional adoption. **RQ2** asks whether attacker-value adoption differs between public-standard facts and asset-specific facts when exposure is held constant. **RQ3** asks how much system-of-record validation reduces out-of-range commitments relative to two generic defenses, and what false-flag cost it introduces when legitimate documents disagree by jurisdiction or interconnection.

The contributions are fourfold: (1) a write-path threat model for low-trust utility-document repositories; (2) GridPoison-100, a benchmark with source-grounded public items, a fictional asset register, and legitimate-conflict controls; (3) a two-stage evaluation that separates retrieval compromise, conditional adoption, and consequence grading; and (4) an accounting of what deterministic system-of-record validation gains and costs under partial coverage and deliberately simplified record scope.

II. RELATED WORK

PoisonedRAG showed that a small number of attacker-controlled texts can steer retrieval-augmented models toward targeted answers [3]. Subsequent work has examined retrieval, database, and generator factors that influence poisoning exposure and attack success [4]. Blocker documents can also jam RAG systems into declining to answer [5]; the non-commitment observed here is related but arises from the assistant’s own conflict-abstention rule rather than an attacker objective. Our commit-required follow-up directly tests that distinction by removing permission to abstain. Because those effects are already known, retrieval depth is a secondary factor here rather than a novelty claim.

RobustRAG introduced an isolate-then-aggregate strategy for retrieval corruption [6]; we use this family as a generic

defense baseline rather than presenting it as new. ClashEval quantified conflicts between parametric knowledge and retrieved evidence [7]. Controlled knowledge-conflict work shows that model behavior can change sharply with conflicting external evidence [8]; recent conflict-aware RAG evaluation also studies abstention under contradictory retrieval [9]. These results motivate our paired-twin versus non-paired-support Stage C control and commit-required manipulation. Other work has shown that evidence phrasing, publication time, and metadata can affect model preferences [10], [11]. These findings motivate our vendor-bulletin condition but also rule out treating authority framing as a primary contribution.

LLM risks in power systems have already been discussed and experimentally explored [1], [2]. Our study therefore avoids claims of being the first LLM-security work in the domain. The domain-specific contribution is the joint measurement of silent corruption, prompt-mediated non-commitment, and record-scope false flags on source-grounded utility lookups.

III. THREAT MODEL

The attacker is an authorized but low-trust writer, such as a contractor with write access to an indexed collaboration repository, or the sender of a spoofed technical bulletin that is subsequently indexed. The attacker may add one to three new documents but cannot edit or delete existing documents. They cannot modify the structured asset/settings record, EMS/SCADA, protection devices, model weights, or the user’s query. The attacker knows the target asset or topic and can read existing documents on it, but does not know the exact query wording, embedding model, or retrieval results.

The attack goal is targeted answer-integrity corruption: a legitimate lookup should commit to an attacker-chosen numeric value. The primary poison is a factual statement rather than an instruction to the model, separating the main condition from indirect prompt injection. One stress paraphrase inherits the directive wording of the clean training-card template but changes only the value. Confidentiality, model-weight poisoning, multi-turn social engineering, and physical control-loop attacks are outside scope. Denial of service is not an attack goal, but we report loss of correct answers because it is the dominant observed effect.

IV. GRIDPOISON-100

GridPoison-100 contains 100 lookup items divided into three tiers. Tier 1 contains 35 public normative facts drawn from Massachusetts DER settings and return-to-service material, OSHA 29 CFR 1910.269 Tables R-6/R-7, and NERC PRC-024-3 interconnection boundaries [12], [13], [14], [15]. The OSHA items state voltage range, phase-to-ground or phase-to-phase exposure, and worksite elevation at or below 900 m. FERC approved PRC-024-4 and PRC-029-1 in 2025, while NERC’s catalog accessed Oct. 10, 2026 still lists PRC-024-3 as mandatory and the newer standards as subject to future enforcement; the benchmark therefore freezes the cited PRC-024-3 version and treats standard version and jurisdiction

as part of scope [16], [17]. Each public item records a source URL, source locator, units, applicability conditions, and an allowable interval.

Tier 2 contains 45 asset-specific facts from a declared fictional utility register. Asset identifiers and values are synthetic by construction, preventing a model from knowing them from pretraining. The register covers transformer emergency ratings, breaker current ratings, feeder thermal limits, BESS continuous discharge limits, and cable ampacity values. For Tier 2, the register is the authoritative record by construction; no claim is made that these values describe a real network.

Tier 3 contains 20 legitimate-conflict items. Two documents can disagree because of jurisdiction (Massachusetts versus IEEE 1547 defaults), NERC interconnection, or table scope (Quebec versus common PRC-024-3 tables), while the query explicitly names which context applies. These items are not poisoned; they measure whether a defense incorrectly flags a correct context-specific answer.

The clean corpus contains 400 short template-generated documents: three support documents per poisonable item in three styles (reference note, lookup summary, and training card), 40 legitimate-conflict documents, and 120 on-topic distractors; each distractor restates another item’s value under a different topic. Documents carry item and asset identifiers, so the corpus is cleaner and more explicit than a real utility repository. Each poison forms a minimal pair with one support document and changes only the critical value, making the conflict maximally visible. The main poison moves the value outside the encoded benchmark interval; a wrong-but-in-range variant separates factual incorrectness from interval violation, and a dated vendor-bulletin form tests authority framing.

V. EXPERIMENTAL DESIGN

A. Reference conditions and models

B0 is closed-book generation, B1 is clean RAG, and B2 is poisoned RAG with the same base prompt. The base prompt requests one committed numeric value when evidence is unambiguous and `VALUE: UNKNOWN` when supplied sources are inconsistent; B2 therefore already permits abstention on conflict, while D1 adds an explicit warning that retrieved sources may be falsified. Retrieval uses sentence-transformers/all-MiniLM-L6-v2 embeddings with cosine similarity and one document per chunk. The evaluated endpoints are Qwen3-8B (qwen/qwen3-8b), Mistral Small 3.2 24B (mistralai/mistral-small-3.2-24b-instruct), and GPT-5.6 Luna (openai/gpt-5.6-luna), called through OpenRouter at temperature zero with `max_tokens=256` and provider-default reasoning settings. Provider token accounting did not enforce that request as a hard reasoning bound for Qwen, while GPT-5.6 Luna reasoning consumed the requested budget. Provider routing was not pinned; resolved model/provider metadata, response ID, token usage, and timestamp were logged per response.

B. Stage A: retrieval exposure

RQ1 uses natural dense retrieval. The lower-strength condition inserts one out-of-range poison; the stress condition inserts three distinct poison paraphrases. Poison@ k is reported separately from generation behavior. A retrieval-only pre-generation calibration found that one-document exposure was strongly imbalanced between Tier 1 and Tier 2, which would confound a direct cross-tier adoption comparison. This exposure gap is a property of this corpus and retriever, not evidence that asset-specific facts resist poisoning.

C. Stage B: forced exposure

RQ2 therefore uses a fixed five-passage context rather than natural retrieval. Each item receives three clean same-item support passages, one deterministic on-topic distractor, and exactly one poison. The poison is counterbalanced in the first or last position. This deliberately evidence-rich design holds exposure and context length constant while comparing public-standard and asset-specific facts; it is not intended to represent natural attack strength.

D. Defenses

D1 is a cautious prompt instructing the model to identify unresolved source conflicts. D2 isolates each retrieved passage, obtains one value per passage with the cautious prompt, ignores non-numeric isolated outputs when voting, and returns the plurality numeric value; no numeric votes or a tie yield UNKNOWN. It is a simplified instance of the isolate-then-aggregate family [6], not RobustRAG’s certified aggregation. D3 post-processes the same B2 stress outputs using deterministic system-of-record validation: if the item’s record is covered and the committed value differs from it, the answer is replaced by a verification flag. D3 is given the applicable record key, so query-to-record linking errors are not modeled. For 78 of the 80 poisonable items the record equals the benchmark target; the exceptions are two Massachusetts undervoltage settings for which the simplified record stores IEEE 1547-2018 Category II defaults [18], [14]. For 13 of the 20 Tier 3 items the record stores a generic alternative rather than the applicable contextual variant. D3 therefore approximates an answer-key check on poisonable items; its purpose here is to quantify an upper-bound filtering benefit and the cost of record-scope mismatch, not to propose a new defense algorithm. Coverage at nominal 40%, 70%, and 100% is fixed deterministically by SHA-256 hashing item identifiers, realizing 30, 57, and 80 of 80 poisonable items and 6, 14, and 20 of 20 Tier 3 items.

RQ3 uses the three-poison $k = 5$ stress condition to avoid a retrieval floor when comparing defenses. The paraphrases rarely co-rank, so the stress condition mainly raises exposure rather than creating a poison majority. Secondary one-factor deviations examine a dated vendor bulletin, a wrong-but-in-range value, and $k = 2$. No full factorial sweep is performed.

VI. METRICS AND STATISTICAL ANALYSIS

The raw logs retain labels for correct (C), attacker-value adoption (A), another committed value (W), conflict flag (F), and refusal (R). For reported metrics, numeric C/A/W grading is deterministic from the required VALUE: line and item tolerance; all 100 benchmark items use zero numeric tolerance. Every response without a parsed numeric value is pooled as non-commitment (N); we do not interpret the parser-level F/R split.

Let E_t indicate whether a poisoned document occurs in the retrieved top- k context. We report Poison@ k , attack success rate (ASR), conditional adoption rate (CAR), correct-response rate (CRR), non-commitment rate (NCR), and benchmark-violation rate (BVR):

$$\text{CAR} = P(A \mid E_t = 1), \quad (1)$$

$$\text{BVR} = \frac{1}{|T|} \sum_t \mathbb{1}[\hat{v}_t \notin [L_{i(t)}, U_{i(t)}]]. \quad (2)$$

BVR counts only committed numeric values; non-committal outputs are non-violations by definition, so BVR is reported together with CRR. It is a benchmark consequence metric, not a physical grid-safety measure.

The item is the unit of analysis and the two query phrasings are repeated observations. Defense differences receive paired 95% item-cluster bootstrap intervals with 2,000 resamples; natural-retrieval CAR intervals also resample item IDs as clusters, and zero-event Stage-B adoption cells receive exact Clopper–Pearson bounds at the item level. The locked primary RQ3 comparison is D3 at 70% coverage versus D1 and D2. Paired comparisons use item-level exact McNemar tests, with an item counted as violating only when both phrasings violate, and Holm correction over the six primary comparisons; comparisons with B2 are secondary and descriptive. Tier 3 BFFR is conditioned on clean-RAG correctness so D3 is charged only when it flags an otherwise-correct context-specific answer. This conditioning is a post-hoc refinement; unconditional BFFR at full coverage was 62.5% for every model. Code, benchmark data, prompts, exact run metadata and timestamps, and the machine-readable methods lock are public at <https://github.com/NTinkicht/Research-GridPoisonRAG>; the final pre-generation alignment is recorded at commit `3de75c6`, and the prospective follow-up lock at `c598f44b`.

VII. RESULTS

Both pre-specified primary endpoints were null: Stage-B attacker-value adoption was 0% in both tiers (RQ2), and no D3@70 comparison with D1 or D2 was significant (RQ3). The exposed-trial breakdown of natural retrieval below was added after these outcomes were known and is descriptive; $k = 2$ was pre-specified as a secondary one-factor deviation. Clean-RAG CRR was 87.8% on Tier 2 (pre-specified interpretation gate: 80%, defined only for Tier 2) and 75.7–78.6% on Tier 1.

A. Natural-retrieval behavior (RQ1)

Table I reports retrieval exposure and generation outcomes. Exposure is deterministic for the shared retriever; CRR and CAR are ranges across models. Adding two poison paragraphs left Tier 1 exposure unchanged at 70.0% but raised Tier 2 exposure from 3.3% to 76.7%; the stress condition therefore increased exposure only in Tier 2. At $k = 5$, one poison was retrieved in 49 Tier 1 trials per model; only 0–3 remained correct and direct attacker adoption was 2/49 (4.1%; item-cluster 95% CI 0.0–10.6). Of those 49 exposed trials, 44–47 were non-committal and 0 produced another wrong numeric value. The same exposure keys had been correct under clean $k = 5$ in 48–49/49 trials, so exposure was concentrated on otherwise-answerable queries. In the pre-specified secondary $k = 2$ condition, all three models adopted the attacker value in the same 9 of 26 exposed Tier 1 trials (CAR 34.6%; item-cluster 95% CI 17.9–54.5). No clean $k = 2$ generation arm was run; among the 44 unexposed Tier 1 trials, each model was correct on 10, so the $k = 2$ CRR row is not a clean- $k = 5$ comparison. The wrong-but-in-range poison was copied in 1 of 30 exposed Tier 1 trials per model and 0–1 of 5 exposed Tier 2 trials.

B. Fixed-context exposure (RQ2)

In the evidence-rich fixed context (three clean same-item supports, one distractor, one poison), attacker-value adoption and BVR were 0% in every model/tier/position cell. With zero events, the exact 95% upper bounds on per-item adoption are 10.0% for Tier 1 ($n = 35$) and 7.9% for Tier 2 ($n = 45$). Table II reports correct-answer retention. Every poison-last trial produced no numeric commitment. Because the base prompt explicitly permits `VALUE: UNKNOWN` when evidence is inconsistent, this is a conflict-handling result rather than evidence of inherent poisoning robustness.

C. Record-check accounting (RQ3)

The locked primary comparison is D3 at 70% coverage versus D1 and D2. Table III reports paired BVR reduction in percentage points; positive values favor D3. Against B2, which uses the same conflict-abstaining base prompt, D3@70 removed 6/7, 5/6, and 6/8 violating trials for Mistral, Qwen, and GPT-5.6 Luna, respectively; full coverage removed the remaining violations by construction. D2 increased Qwen BVR by 3.1 points relative to B2. Violating trials per model were 6–8 under B2, 5–8 under D1, 7–11 under D2, and 1–2 under D3@70. The primary McNemar comparisons had at most 3 discordant items and all six Holm-adjusted tests were non-significant ($p_{\text{Holm}} = 1.0$). The defense comparisons are descriptive. With at most 3 discordant items, the smallest attainable exact two-sided p was 0.25 before adjustment, so the locked comparisons could not reach significance at these violation rates. Bootstrap intervals can exclude zero while McNemar remains non-significant because the bootstrap estimates trial-level mean differences while clustering by item, whereas McNemar dichotomizes an item only when both phrasings violate; sparse discordance makes the exact test coarse. Because

BVR counts non-commitment as a non-violation, note that across the stress-condition defenses CRR was 10.6–17.5% and non-commitment was 77.5–88.8%; D3@70 lowered CRR by 0.6–1.2 points relative to B2. In a post-hoc sensitivity analysis, counting an item as violating when either phrasing violates also leaves all six comparisons non-significant (minimum $p_{\text{Holm}} = 0.094$).

D. Legitimate-conflict cost

Clean RAG answered 35–39 of 40 legitimate-conflict trials correctly. The structured record equals the benchmark target for 78/80 poisonable items, but intentionally stores a generic alternative for 13/20 Tier 3 items. BFFR therefore measures a controlled record-scope gap and is conditioned on clean-RAG correctness (a post-hoc refinement; unconditional BFFR at full coverage was 62.5% for every model). At 70% coverage, conditional BFFR was 43.6–45.7%; at full coverage it was 64.1–65.7%. The cost is therefore driven by record scope, not only by model behavior.

E. Prospective Follow-up and Exploratory Backup

After the original Qwen/Mistral/Luna results were frozen, we locked four follow-up checks before running them. Qwen’s follow-up runs did not complete because of repeated provider-layer failures; the completed Mistral and Luna runs are reported as locked follow-ups. Gemini 2.5 Flash-Lite was then run as an exploratory backup with its own matched B2 baseline; it is not a pre-specified model, and no locked verdict depends on it.

Design. (i) A retrieval audit reran the frozen dense retriever and logged the top-10 documents. (ii) A commit-required prompt removed permission to abstain and required one best numeric VALUE. It reused the one-poison $k = 5$ setting; CAR is evaluated on the same 49 exposed Tier 1 trials per locked model. Confidence intervals resample item IDs, and paired inference collapses repeated phrasings to the item, positive if any originally exposed phrasing adopts. (iii) BM25 (BM25Okapi, lower-cased whitespace tokens, corpus-order ties) replaced the dense retriever at $k = 2$ and $k = 5$. (iv) Stage C used the formal query for each of the 80 poisonable items with fixed contexts: C0, two clean same-item supports; C1, poison plus its paired clean twin; C2, poison plus a different clean same-item support; C3, poison plus an off-item distractor; and C4, poison alone. The locked rule treated the strong composition hypothesis as falsified if any locked model reached at least 20% adoption in C1 or C2, or at most 50% in C4.

Results. The audit reproduced all 640 original retrieval contexts (1920 generation rows across three models) with no poison-rank mismatch (one-poison $k = 5$, three-poison $k = 5$, bulletin $k = 5$, and one-poison $k = 2$). Across both tiers at $k = 2$, 9 exposed trials had no clean same-item passage and all 9 adopted; 20 exposed trials had at least one clean same-item passage and none adopted. Because the adoption outcomes were known before this audit, this is a consistency check rather than an independent test. Requiring commitment

TABLE I

NATURAL-RETRIEVAL OUTCOMES (%; $k = 5$ UNLESS NOTED). CAR IS CONDITIONAL ON EXPOSURE; EXPOSED TIER 1/TIER 2 TRIALS PER MODEL ARE 49/3 (1 POISON), 49/69 (3 POISONS), 47/7 (BULLETIN), AND 26/3 ($k = 2$). THE CLEAN ROW IS $k = 5$; NO CLEAN $k = 2$ GENERATION ARM WAS RUN.

Condition	Tier 1 (public)			Tier 2 (asset)		
	Exp.	CRR	CAR	Exp.	CRR	CAR
Clean RAG	—	75.7–78.6%	—	—	87.8%	—
1 poison	70.0%	8.6–12.9%	4.1%	3.3%	84.4%	0.0%
3 poisons	70.0%	7.1–15.7%	4.1%	76.7%	14.4%	4.3–5.8%
1 bulletin	67.1%	20.0–22.9%	6.4–8.5%	7.8%	80.0%	0.0–14.3%
1 poison, $k = 2$	37.1%	14.3%	34.6%	3.3%	62.2%	0.0%

TABLE II
CORRECT-ANSWER RATE UNDER FIXED EXPOSURE (%).

Model	First/T1	First/T2	Last/T1	Last/T2
Mistral Small 3.2	8.6%	97.8%	0.0%	0.0%
Qwen3-8B	42.9%	57.8%	0.0%	0.0%
GPT-5.6 Luna	74.3%	28.9%	0.0%	0.0%

raised adoption from 2/49 to 24/49 for Mistral (49.0%; item-cluster 95% CI 33.3–64.4; item-level exact $p = 3.05 \times 10^{-5}$, $n = 31$ items) and from 2/49 to 8/49 for Luna (16.3%; 7.5–26.5; item-level exact $p = 0.0312$, $n = 31$ items). The pre-stated point-estimate bands were $\leq 10\%$ model resistance, > 10 to $< 25\%$ partial mediation, and $\geq 25\%$ material mediation; Mistral is material and Luna partial. Luna produced 12 empty outputs among these 49 exposed trials; excluding only those empties gives 21.6% adoption. The commit-prompt clean controls retained 83.8% CRR for Mistral and 86.9% for Luna (exploratory Gemini: 84.4%). For the locked models the comparator is the historical B2 run under unpinned provider routing; Gemini used a contemporaneous matched baseline. An empty-output audit found none in Mistral or Qwen B2_P1 or either Stage-B position. Luna had 1/160 empties in B2_P1, 4/80 with poison first and 0/80 with poison last. In Luna Stage C, C2 and C3 each had 2/80 empties and C0/C1/C4 had none; Mistral had none. Qwen Stage C is unavailable because its follow-up failed. Stage C falsified the pre-stated strong composition hypothesis: Mistral adopted in 17/80 C2 trials (21.2%; exact 95% CI 12.9–31.8), one trial above the threshold, while Luna (15/80) met every threshold. Both locked models adopted in 0/80 C1 and 80/80 C4 trials. In C1 the paired twin protected integrity by producing non-commitment rather than correct answers: both locked models were 0/80 correct. The no-poison C0 control was 0/80 adoption and 80/80 correct for every model. Table V reports all C1–C4 adoption counts.

Under BM25, the poison and its paired clean twin had exactly equal scores in 160/160 queries. Exposure was therefore highly tie-order dependent: under the locked clean-first corpus order, the poison appeared in 2/160 queries at $k = 2$ and 38/160 at $k = 5$; placing the poison first among equal scores changes those counts to 112/160 and 124/160, respectively. BM25 is therefore a tie-order sensitivity, not evidence of lexical robustness. *Post-hoc benchmark consequence mapping*. A

post-hoc benchmark-consequence mapping decomposed each OSHA MAD into its electrical component and inadvertent-movement allowance. Of the 18 MAD poisons, 13 consumed only part of the allowance, leaving 0.03–0.45 m beyond the electrical component, and 5 fell inside the electrical component by 0.01–0.17 m. All 7 PRC-024-3 poisons shortened the benchmark no-trip time by 60%. All 8 Massachusetts DER trip-threshold poisons moved the encoded threshold farther from nominal, widening the benchmark no-trip region. The remaining 2 public exact-value items were return-to-service values and are not assigned a directional consequence class here. All 45 fictional Tier-2 poisons exceeded the record limit by 20%. 2 of these electrical-component cases (T1-025, T1-027) were among the nine $k = 2$ adopted trials. These are benchmark consequence classes, not professional-engineering or physical-safety determinations. *Post-hoc coverage what-if*. Severity-prioritized D3 gave no consistent benefit: relative to hash coverage, it changed violating trials per model by -1 to +0 of 160 at 40% and by +1 to +2 at 70%, with Tier-3 coverage held fixed.

VIII. DISCUSSION

Natural retrieval shows how poisoning manifests under a conflict-abstaining prompt. At $k = 5$, a retrieved poison rarely transferred its value, but exposed Tier 1 trials usually lost the correct answer and became non-committal. Operationally these are different failures: non-commitment is an availability loss that would require an external handling policy, whereas attacker-value adoption is a silent integrity failure. The commit-required control supports material prompt mediation for Mistral and partial mediation for Luna: without permission to abstain, 24 of 49 and 8 of 49 exposed trials adopted the attacker value, respectively. In the pre-specified secondary $k = 2$ condition, all three original models adopted the attacker value on the same nine exposed trials, and the reproduced contexts show that none of the nine contained a clean same-item passage. The forced-composition control supports a narrower mechanism than that association suggests: under the base prompt, the poison’s paired clean twin turned adoption into abstention (0/80 adoptions and 0/80 correct answers for both locked models), whereas a different clean support left 15–17/80 adoptions in the locked models and 63/80 in the exploratory backup. Context composition therefore shapes adoption strongly, but whether a clean copy protects depends on which copy is retrieved and on the model. Stage C used

TABLE III

PRIMARY RQ3 COMPARISON AND VIOLATION COUNTS. POSITIVE PP VALUES FAVOR D3. BRACKETS ARE DESCRIPTIVE 95% ITEM-CLUSTER BOOTSTRAP INTERVALS OF TRIAL-LEVEL BVR DIFFERENCES; MCNEMAR TESTS COUNT AN ITEM AS VIOLATING ONLY WHEN BOTH PHRASINGS VIOLATE.

Model	D3@70 vs D1 (pp)	D3@70 vs D2 (pp)	B2 → D3@70	D3@100
Mistral Small 3.2	2.5 [-0.6, 6.2]	3.8 [0.6, 8.1]	7 → 1	0
Qwen3-8B	3.1 [0.6, 6.9]	6.2 [1.9, 11.2]	6 → 1	0
GPT-5.6 Luna	3.8 [0.6, 7.5]	3.8 [0.0, 8.8]	8 → 2	0

TABLE IV

CONDITIONAL BENIGN-CONFLICT FALSE-FLAG RATE (BFFR).

Model	40%	70%	100%
Mistral Small 3.2	17.9%	43.6%	64.1%
Qwen3-8B	17.1%	45.7%	65.7%
GPT-5.6 Luna	17.9%	43.6%	64.1%

TABLE V

FOLLOW-UP CONTROLS (COUNTS). COMMIT: ATTACKER ADOPTIONS AMONG 49 EXPOSED TIER 1 TRIALS UNDER THE COMMIT-REQUIRED PROMPT (BASE PROMPT: 2, 2, AND 3). BM25: ADOPTIONS AMONG TRIALS EXPOSED AT $k = 5$ UNDER THE LOCKED CORPUS-ORDER TIE RULE. C1–C4: ADOPTIONS OUT OF 80 FORCED-CONTEXT TRIALS. GEMINI 2.5 FLASH-LITE IS EXPLORATORY.

Model	Commit	BM25	C1	C2	C3	C4
Mistral Small 3.2	24/49	2/38	0	17	53	80
GPT-5.6 Luna	8/49	0/38	0	15	73	80
Gemini (expl.)	21/49	12/38	1	63	78	80

only the base prompt, so its paired-twin result is consistent with conflict-abstention behavior but does not independently identify the prompt instruction as the mechanism.

RQ2 was inconclusive rather than evidence of robustness: forced-exposure adoption was at floor in both tiers. The evidence-rich Stage-B context contains three clean same-item supports against one poison, and the base prompt explicitly permits abstention on unresolved conflict. Closed-book Tier 1 CRR was 0–2.9%, but GPT-5.6 Luna returned empty output in 41 of 70 closed-book Tier 1 trials under the requested token budget, so its parametric prior was not adequately measured.

D3 measures record coverage rather than language-model behavior. Because it is given the applicable record key and the record equals the target for 78/80 poisonable items, full coverage removes observed violations by construction. Under independent random coverage c , expected residual violations scale with the uncovered fraction $(1 - c)$; our deterministic hash is one realized coverage draw, with 57/80 poisonable items covered at the nominal 70% level. The locked D3-vs-D1/D2 comparisons involve few violating items and remain non-significant, so they are descriptive.

The Tier 3 control isolates record-scope cost, although its magnitude is partly designed by the 13/20 generic-record construction. Clean RAG answered 35–39 of 40 scoped-conflict trials correctly, showing that the models usually resolved legitimate disagreements when the query named the applicable jurisdiction or interconnection. A generic SoR then flags many

of those correct answers because those 13 records intentionally omit the applicable contextual variant. These results support scoping record authority to the applicable jurisdiction or interconnection; escalation policy was not varied.

IX. CONCLUSION

GridPoison-100 separates retrieval exposure, attacker-value adoption, non-commitment, and benchmark consequence in utility-operations RAG. Under a conflict-abstaining prompt, one retrieved poison at $k = 5$ usually converted a correct lookup into non-commitment rather than silent corruption; the locked commit-required control supported material prompt mediation for Mistral and partial mediation for Luna, with exposed-trial CAR of 49.0% and 16.3%. RQ2 remained inconclusive at floor: fixed-context attacker adoption was 0% in both tiers under the evidence-rich abstention-capable design. Adoption tracked context composition: 0/80 in both locked models when the poison’s paired clean twin was present and 80/80 for a lone poison, but a different clean support did not reliably prevent it, so the pre-stated strong composition hypothesis was falsified. At the locked 70% coverage, the keyed structured-record check removed 5–6 of 6–8 B2 violating trials across models; full coverage removed the remaining violations by construction, while record-scope mismatch could flag correct context-specific answers. The design implication these data support directly is to scope structured records to the jurisdiction and interconnection in which they are authoritative; whether redundant approved evidence limits silent corruption in real repositories remains untested.

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